

Arman Hassan

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EDUCATION

University of North Carolina at Chapel Hill

Chapel Hill, NC

Bachelor of Science in Computer Science — Minor in Data Science

August 2023 – May 2027

- Coursework: Data Structures, Algorithms, Systems, Computer Organization, Linear Algebra, Models of Computation, Machine Learning, Intro to AI, Mobile Computing, Hardware Security, Data Management

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, C++, SQL, HTML/CSS

Frameworks: React, React Native, Next.js, Node.js, Expo, FastAPI, Playwright, Tailwind CSS

Cloud & DevOps: AWS (Bedrock, S3, IAM), Oracle Cloud, Vercel, Docker, Git, GitHub

Databases: PostgreSQL, pgvector, Redis, Amazon S3 Vectors

AI & Data: OpenAI API, Claude, RAG, LLM Agents, scikit-learn, pandas, NumPy

RELEVANT EXPERIENCE

Timing

Remote

Software Engineering Intern

January 2026 – May 2026

- Built an **OpenAI-powered agent** using function calling and structured outputs to rank contacts from interaction history, reducing manual prioritization by **60%**
- Implemented a **RAG pipeline** with **PostgreSQL** and **pgvector** across **1,000+ contact records** to retrieve relevant context and generate personalized outreach drafts
- Reduced API latency from **450ms** to **under 90ms** by adding composite PostgreSQL indexes and **Redis caching**

Introduction to Programming

Chapel Hill, NC

Teaching Assistant

August 2025 – December 2025

- Supported **200+ students** through weekly Python labs and office hours, diagnosing bugs and teaching control flow, functions, OOP, runtime analysis, and memory diagrams
- Reviewed and graded programming assignments, provided code-level feedback, and coordinated with course staff to address recurring student misconceptions

Vogro

Remote

Software Engineering Intern

May 2025 – August 2025

- Developed **Python automation tools** with input validation, structured logging, and exception handling, reducing manual data-processing time by **40%**
- Built REST API integrations with **FastAPI** and **Node.js** connecting a React frontend, backend services, and third-party tools across **3+ releases** serving **500+ users**
- Reduced page load time by **20%** by eliminating redundant API requests and refactoring React components for improved performance on lower-end devices

PROJECTS

Jev 2048 | [Live Demo](#)

Chapel Hill, NC

Personal Project | Next.js, TypeScript, TypeSafe Jev, C++, Python, Oracle Cloud, Vercel

September 2026

- Built a controlled experiment implementing the general-purpose **TypeSafe Jev classifier** as a sequential decision policy to autonomously play 2048
- Benchmarked **five policies** using raw boards, structural features, expectimax, and pretrained n-tuple values
- Deployed a Vercel UI and authenticated Oracle Cloud API serving an MIT-licensed, **67.1M-weight C++ model**
- Two-seed pilot: mean score rose from **796** raw to **10,770** with features and **34,276** with expectimax guidance
- Processed **77,928 API requests** and **133.1M tokens** for **\$5.45**, demonstrating low-cost classification at scale

Degree Planner AI | [Live Demo](#)

Chapel Hill, NC

Personal Project | Next.js, TypeScript, AWS Bedrock, S3 Vectors, Claude, Vercel

August 2026

- Built and deployed a degree-planning assistant covering 50+ UNC majors and minors with cited answers
- Parsed catalog pages into structured records and indexed Titan embeddings in **S3 Vectors**
- Built a **RAG pipeline** with **AWS Bedrock** to map completed courses to remaining requirements
- Tested retrieval scores and added Claude routing to improve relevance and reject unrelated questions
- Added rate limits, access-code protection, and an automated AWS budget cap to control usage

Arcade-Style Space Battle Game

Chapel Hill, NC

Hack110 Hackathon — **1st Place** | Python, Pygame, OOP, Git

April 2024

- Won **1st place** at Hack110 by shipping a full Pygame arcade shooter end-to-end in under 24 hours.
- Built core game systems including collision detection, enemy spawning, and progressive difficulty scaling.